

$$A_v = G_m R_{out}$$

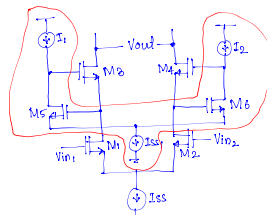
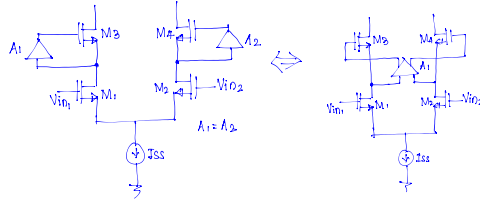
$$G_m = g_{m1}$$

$$R_{out} = (1 + A_{ux}) g_{m2} r_{o2} r_{o1}$$

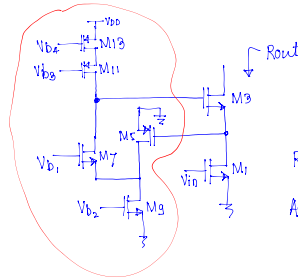
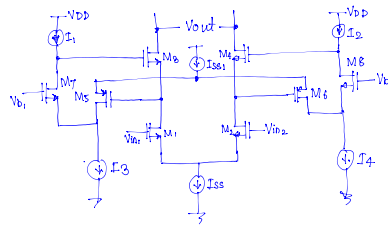
$$A_{ux} = G_{m_{aux}} R_{out_{aux}}$$

$$= g_{m3} r_{o3} r_{o2}$$

$$\therefore A_v = g_{m1} [1 + g_{m2} g_{m3} r_{o2} r_{o3}] g_{m2} r_{o2} r_{o1}$$



$$V_{out,min} = V_{SS1} + V_{GS5} + V_{OV3}$$

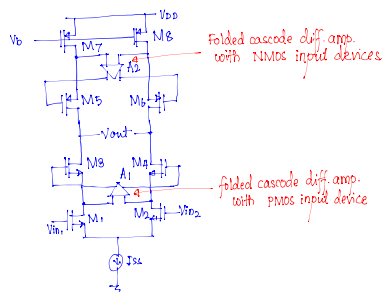


$$R_{out} = (1 + A_{ux}) g_{m2} r_{o2} r_{o1}$$

$$A_{ux} = G_{m_{aux}} R_{out_{aux}}$$

$$= g_{m5} [R_{up_{aux}} || R_{down_{aux}}]$$

$$= g_{m5} [g_{m1} r_{o1} r_{o2} || g_{m7} r_{o7} (r_{o3} || r_{o5})]$$



Common Mode Feedback (CMFB)

